

Daniel Alejandro Silva Rojas

SENIOR SYSTEMS & PERFORMANCE ENGINEER · BACKEND & AI

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PROFESSIONAL SUMMARY

Systems & Performance Engineer with 5+ years of experience leading low-latency, real-time architectures in C++, Rust, and Python. At Ea2technology (Canada) I design industrial Digital Twins for refinery monitoring and full-duplex WebSocket architectures that reduced latency from 200-500ms to 1-10ms. I complement this with computer vision, scientific computing (NumPy/SciPy), and RAG systems, plus full-stack development (Vue3/Nuxt3, FastAPI, Supabase). Academic training (B.Sc. in Computer Science, LUZ-IUTA) completed in parallel with active professional experience since 2021. Founder of DropAudio CCS, an e-commerce platform in production — a part-time personal project alongside my full-time primary role and occasional freelance consulting.

TECHNICAL SKILLS

Languages: Rust, C++, Python, TypeScript, JavaScript (ES6+), SQL

Systems & Real-Time: Tokio, Axum, Rocket, full-duplex WebSockets, gRPC, TCP Sockets, Digital Twin architectures

AI & Scientific Computing: FastAPI, RAG systems, NumPy, SciPy, OpenCV, PyO3, FlowNet / deep learning

Data & Cloud: PostgreSQL, MySQL, Supabase, Redis, MongoDB, AWS (EC2, S3, RDS, Lambda), Docker

Specialties: Low-latency architecture, concurrency, memory safety, industrial telemetry, computer vision

PROFESSIONAL EXPERIENCE

Lead Performance & Systems Engineer

Jul 2021 – Present

Ea2technology — Canada (remote)

- Designed and deployed an industrial Digital Twin system (telemetry) for real-time monitoring of coke drum deformation in a refinery, processing high-frequency sensor data to predict and prevent critical structural failures.
- Designed a low-latency, full-duplex WebSocket architecture for dynamic dashboards, replacing HTTP polling: reduced latency from 200-500ms to 1-10ms and cut bandwidth usage by up to 95%.
- Led the migration of legacy monolithic systems to cloud-native microservices on AWS, optimizing compute-intensive data pipelines with C++ and Rust to ensure memory safety and maximum concurrency under load.
- Led end-to-end technical UI/UX design in Figma, connecting low-level performance with intuitive operator interfaces.

Founder & Software Architect

Jun 2021 – Present

DropAudio CCS — dropaudioccs.com (e-commerce platform in production)

- Designed and launched a complete e-commerce platform (Nuxt 3 SSR + Supabase/PostgreSQL with RLS) to production, hosted on Vercel, consolidating 94+ verified customer reviews.
- Developed a 3-step interactive recommender and technical comparison tool that reduced purchase friction and increased conversion.
- Implemented a custom multi-currency checkout (USDT, Zinli, Pago Móvil) with live BCV/USDT exchange rates and automated Web Push notifications via pg_cron.
- Built a resilient architecture with a backup catalog for outages and technical SEO/GEO (JSON-LD, sitemap, PWA) to ensure operational continuity.

Software Performance Consultant & Algorithm Engineer

Jan 2023 – Present

Freelance / Independent Consulting · Remote — international client Vessel Inspection and Assessment Corp (VIAAC), maritime/oil sector — role: Senior Developer

- Developed high-performance video stabilization engines using optical flow algorithms and matrix interpolation (Python, NumPy, SciPy, OpenCV) for trajectory smoothing via Euclidean transformations.
- Implemented global motion estimation with FlowNet (distilled deep architectures) to correct unstable video, combining traditional mathematics with modern machine learning.
- Designed and deployed TCP Socket and WebSocket communication systems, including custom file-transfer protocols, for high-frequency environments.
- Conducted technology-stack audits to identify memory leaks, concurrency issues, and latency bottlenecks, delivering refactoring roadmaps in optimized Rust, C++, and Python (PyO3).

IT Support Specialist

Jan 2021 – Sep 2021

René Dessés de Venezuela — Caracas, Venezuela

- Maintained equipment, software, and Apache/Linux servers.
- Installed and administered enterprise networks, including Cisco infrastructure.

FEATURED PROJECTS

DropAudio CCS *Nuxt 3 · Supabase · Vercel*

Complete e-commerce platform in production with admin panel, catalog, multi-currency checkout, and interactive audio recommender.

RealtimeVoiceAssistant *Rust · Python · Gemini Live*

Real-time conversational voice assistant with transcription, NLP, and streaming voice synthesis, built on an async Rust runtime (Tokio).

System_Stabilization_Interpolation *Python · PyTorch · OpenCV — Degree Thesis*

Video stabilization on mobile devices using deep learning and mathematical interpolation; architecture optimized for real-time processing with numerical precision control to avoid critical rounding errors.

FlowNet Video Stabilization *Python*

Video stabilization system using deep learning: global motion estimation distilled from optical flow.

extract_rif *FastAPI · PaddleOCR*

HTTP microservice that extracts structured data from RIF documents (SENIAT) from PDF or image via dual-engine OCR.

Socket Communication System *Python*

Client-server communication system over TCP sockets with a custom protocol.

EDUCATION

Bachelor's Degree in Computer Science

Graduation: April 2025

LUZ-IUTA (Universidad del Zulia — University Institute of Technology), Venezuela · Studies: Oct 2023 – Apr 2025 · Program equivalent to a B.Sc. in Computer Science

Associate Degree (TSU) in Information Technology

2018 – 2021

LUZ-IUTA, Venezuela

LANGUAGES

Spanish — Native · English — B2 (Professional; advanced technical reading/writing, interviews, and professional communication)

CERTIFICATIONS

Vue.js 2 Professional Course (Platzi, Jan 2024) · TypeScript: Advanced Types and Functions (Platzi, Jan 2024) · TypeScript Fundamentals (Platzi, Jan 2024) · Web Optimization (Platzi, Aug 2024) · Python: NumPy and Pandas for Data Analysis (Platzi, Aug 2024) · Introduction to Generative AI — Duke University (Coursera, Feb 2025)